

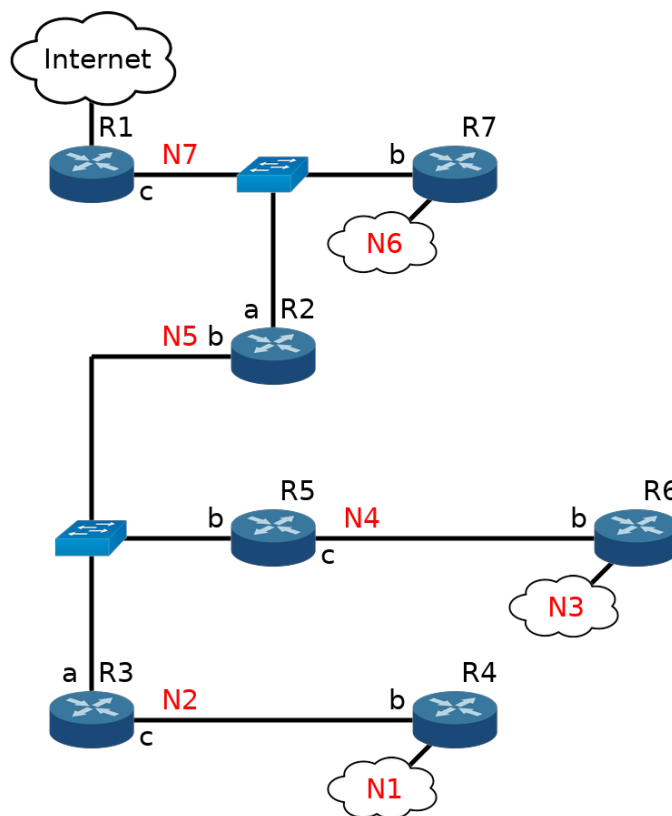
BIE-PSI Test 2 – XX

1. (3 points) The following figure shows the topology of a local network connected to the Internet. The local network consists of several network segments marked as N0, N1, ... (in red). Each router is marked R1, R2, ... and its network interfaces are marked a, b, c, d. Each router has a routing table, which always contains pairs - the destination network and the gateway. The destination network corresponds to the segment names (N1, N2, ...) or the DG value, which means that the destination is the Default Gateway. The second part represents the gateway (or its address) and consists of a combination of the router name and the network interface name separated by a dot, eg: R1.a or R3.d.

Your task is to fill in the routing tables so that all subnets can communicate with the Internet. Assumptions:

- Assume that the R1 router has a working NAT and the Internet connection is guaranteed from this point of view.
- For the R1 router, you do not need to fill in the default gateway record.
- Do not fill directly connected segments into tables.
- Always specify the router address in the direction of the Internet as the default gateway.
- Save work and do not fill in routing tables with paths to segments that are already reachable through the default gateway (the solution may not be optimal, but it must be functional).

Fill in the routing tables of all routers (on the following sheet) so that each record consists of two items - the destination network identifier and the network interface identifier, which serves as a gateway to the destination network.



Fill in the routing tables on the following sheet →

2. The network in the picture (below) contains:

- Routers R1, R2, ...
- Switches S1, S2, ...
- Computers P1, P2, ...

Each computer is assigned:

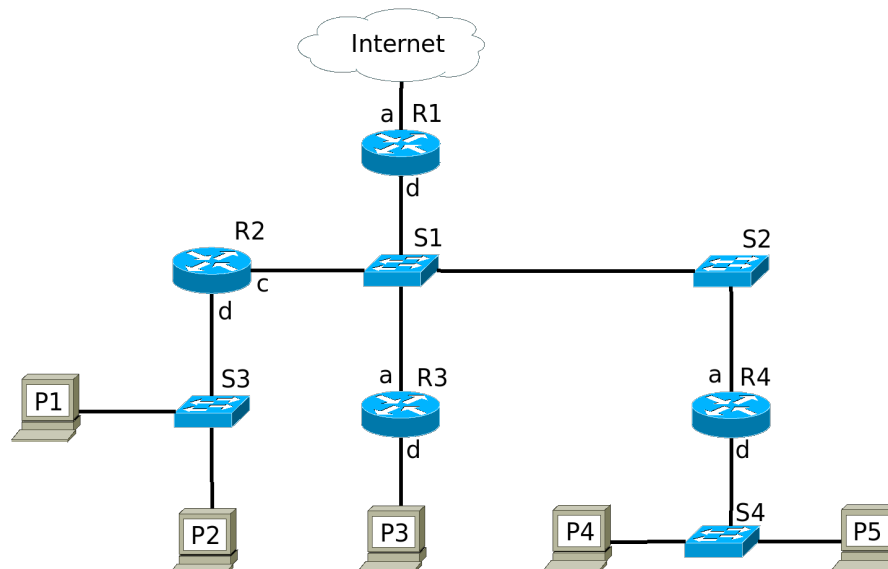
- one IP address, which is referred to for each computer as a combination of "IP" and its name: IPP1, IPP2, ...
- one MAC address, which is referred to for each computer as a combination of "MAC" and its name: MACP1, MACP2, ...

Each router has:

- 2 to 4 network interfaces marked as a, b, c or d
- for each network interface IP address is defined as a combination of "IP", router name and interface name, e.g.: IPR1a, IPR2c, ...
- for each network interface MAC address is defined as a combination of "MAC", router name and interface name, e.g.: MACR1a, MACR2c, ...

Assumptions:

- router R1 performs dynamic NAT between internal network at interface d and external network at interface a
- DNS service is running on the R1 router and is available on its internal network interface
- router R2 performs dynamic NAT between internal network at interface d and external network at interface c
- router R2 performs port forwarding of the SSH port (22) to computer P1
- router R2 performs port forwarding of HTTP port (80) to computer P2
- all network interfaces, except those on the end stations, have an IP address configured statically
- all statically configured IP addresses survive the device restart
- network interfaces on end stations get their configuration via DHCP service, the default gateway is always the IP address of the router towards the Internet and DNS is always the IPR1d address
- each router has a DHCP protocol running for segments in which end stations are located (no DHCP relay)
- each router has correctly filled in routing tables so that the shortest path between individual parts of the network is always guaranteed (except for the part of the network hidden behind R2, which is not in the routing tables of any router except R2)
- routing tables on routers survive device restart
- all routers have the default gateway to the internet filled in correctly towards the internet



On the next sheet, write the answers to the following questions:

- (1 point) P1 sends a message to P5, by how much and where does the TTL value in the IP packet header decrease?
- (3 points) P2 sends a message to the IP address 8.8.8.8 and receives a response. Describe all changes in the source IP addresses in the message and response from network interface P2 to network interface R1a inclusive. Describe the reasons for your answer.
- (5 points) Assume that all devices on the network have just turned on. (All dynamic parameters and tables are empty.) P3 wants to send one ICMP-request (ping) to IPP4. Describe how many frames pass through S1? Describe their sequence, top-level protocol, destination and source MAC addresses, and IP addresses.

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Username: _____

Fill in the routing tables for the routers from task 1) here:

[illegible]

Note: The first column contains the identifier of the destination network and the second column contains identifier of the interface that is the gateway to the destination network.

Answer the questions from task 2) here:

a)

b)

c)

[illegible]

